KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 7 — THE GLOBAL WARNING

Global food markets are no longer stable.Export bans, price spikes, and supply shocks are reshaping food security worldwide. For import dependent countries like Bangladesh, this is not a future risk it is a current vulnerability.

Analysis : Global food prices (2015–2024), export bans, and cross-country food security comparison
Tools : SQL, Power BI
Datasets : global_food_prices.csv (10 rows) export_bans.csv (8 rows) country_comparison.csv (10 rows)

Author: Md. Rafijur Rahman
Data Analyst | SQL | Power BI

Project Repository: https://github.com/rafijurrahman/krishigrid-bangladesh
LinkedIn: linkedin.com/in/mdrafijur

THE GLOBAL WARNING

THE QUESTION

Why must Bangladesh act NOW to reduce food import dependency?

APPROACH

- Analyze global food price volatility (2015–2024) using FAO indices
- Examine real export bans affecting Bangladesh supply chains
- Compare Bangladesh with 10 countries on food security metrics
- Assess relationship between import dependency and economic risk

WHAT WE MEASURE

- Global food price instability
- Frequency and impact of export bans
- Bangladesh's dependency vs other countries
- Food security ranking and structural risk
- Vulnerability to external supply shocks

DATASET OVERVIEW

File 1 : global_food_prices.csv
Rows : 10
Columns: 16
File 2 : export_bans.csv
Rows : 8
Columns: 12
File 3 : country_comparison.csv
Rows : 10
Columns: 17

Key Fields:

- Food price indices (FAO overall, cereal, oil, sugar)
- International commodity prices (rice, palm oil, wheat)
- Export ban duration and price impact (%)
- Import dependency (oil, rice, wheat)
- Food security index score and rank
- Forex reserves and import cover

Source: FAO | World Bank | IMF | Bangladesh Bank | USDA | UNCTAD

SQL Queries

Question
Are global food prices stable or highly volatile over time?

```
SELECT
    year,
    fao_food_price_index,
    fao_cereal_price_index,
    fao_vegetable_oil_price_index,
    international_rice_price_usd_per_ton,
    international_palm_oil_price_usd_per_ton,
    international_wheat_price_usd_per_ton,
    bangladesh_food_inflation_pct,
    major_global_event
FROM global_food_prices
ORDER BY year;
```

Key Insight:

Global food prices are clearly not stable, but highly shock-driven. The FAO food price index rises from around 93 (2015) to a peak of ~144 (2022), while vegetable oil prices show even sharper spikes (~85 → 171 → 125), indicating extreme volatility rather than steady growth. Commodity prices follow the same pattern, with palm oil jumping from ~620 USD/ton to 1430 USD/ton, and wheat from ~202 to 390 USD/ton during crisis periods. More importantly, these spikes directly align with global events COVID-19, Russia-Ukraine war, and export bans and are immediately reflected in Bangladesh's food inflation, which rises to ~11% in 2022. This confirms that global food markets are reactive and unpredictable, meaning import-dependent countries like Bangladesh are exposed to sudden external shocks rather than gradual, manageable changes.

Question
How do export bans from major countries impact Bangladesh's food supply and prices?

```
SELECT
    country,
    product_affected,
    ban_type,
    start_date,
    end_date,
    duration_months,
    bangladesh_import_share_from_this_country_pct,
    estimated_price_increase_in_bangladesh_pct,
    reason
FROM export_bans
ORDER BY start_date;
```

Key Insight:

The data shows that Bangladesh's food system is heavily exposed to decisions made by a few exporting countries, with dependency levels reaching 25% to 72% for key commodities like rice and wheat. When these countries impose export bans, the impact is immediate, triggering sharp price increases ranging from 28% to 180%, especially for essential items like onions and edible oils. Even short-term disruptions, such as Indonesia's 2-month palm oil ban, caused a 42% price surge, while longer bans from India and Ukraine extend supply constraints for 18–34 months. These actions are driven by internal priorities of exporting nations, meaning Bangladesh has no control over them.
Overall, this confirms a structural vulnerability: Bangladesh's food security is directly influenced by external policy decisions, making import dependency a serious economic and strategic risk.

SQL Queries

Question
How does Bangladesh compare with other countries in terms of food import dependency and overall food security?

```
SELECT
    country,
    edible_oil_import_dependency_pct,
    rice_import_dependency_pct,
    wheat_import_dependency_pct,
    overall_food_import_bill_usd_billion,
    global_food_security_index_score,
    global_food_security_index_rank,
    forex_reserve_usd_billion,
    forex_reserve_months_of_import_cover
FROM country_comparison
ORDER BY global_food_security_index_rank;
```

Key Insight:

The comparison clearly shows that Bangladesh is among the most vulnerable countries in terms of food security, driven by extremely high import dependency and weak financial buffers. With 90% dependency on edible oil and 80% on wheat, Bangladesh stands significantly higher than most peers, while countries like India and Thailand maintain near self-sufficiency in key staples. At the same time, Bangladesh's food security rank (84) is considerably lower than stronger performers like Thailand (39) and Vietnam (55), indicating structural weaknesses rather than temporary issues.

What makes this more critical is the limited financial protection — Bangladesh holds only about $19.5 billion in forex reserves, covering just ~2.2 months of imports, compared to countries like India (11.5 months) or Thailand (7 months). This means Bangladesh has far less ability to absorb global price shocks or supply disruptions. In contrast, countries with either lower dependency or stronger reserves are significantly more resilient.

Overall, the data confirms that Bangladesh faces a dual risk: high dependency + low buffer capacity, making it one of the most exposed countries in any global food crisis. This is not just an agricultural issue, but a structural economic vulnerability that requires immediate and systemic intervention.

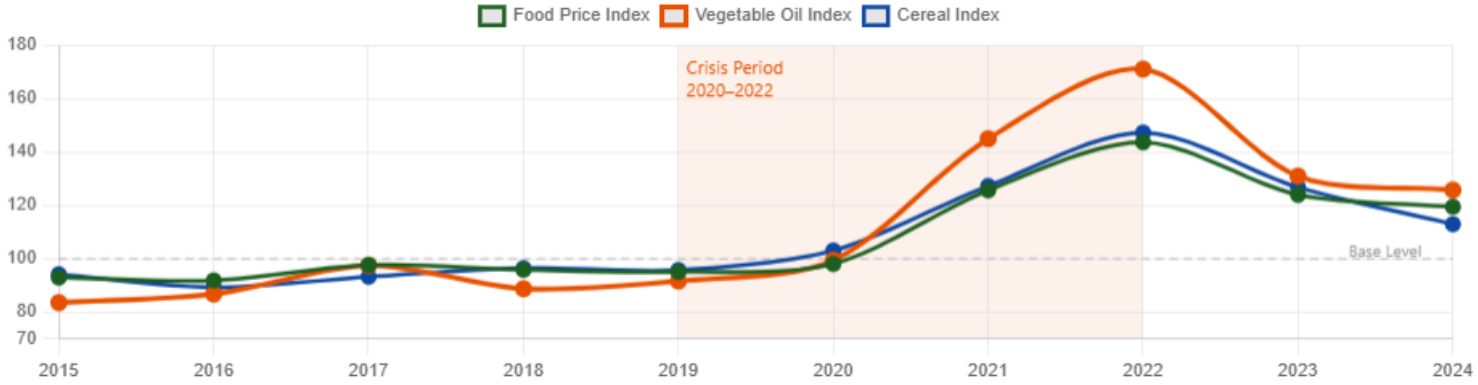
Peak FAO Vegetable Oil Index
171.1
(2014-16 = 100)

Peak Palm Oil Price
1,430
USD/Ton

Total Export Bans Recorded
8
incidents tracked

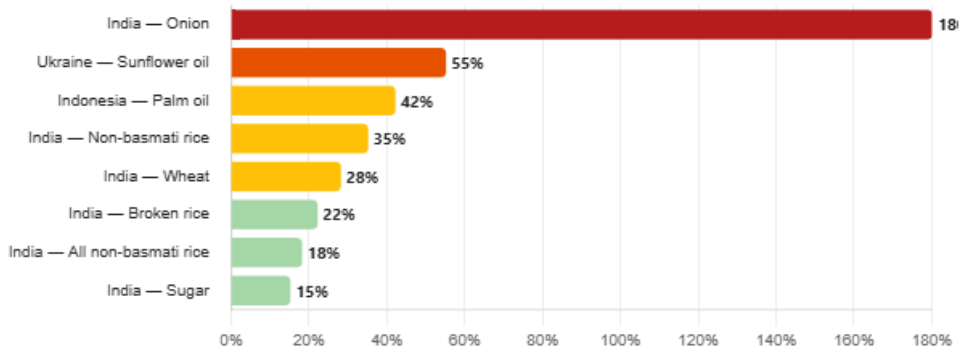
Max Price Spike in Bangladesh
180
% price increase

Global Food Price Indices 2015–2024 (Base 2014-16 = 100)



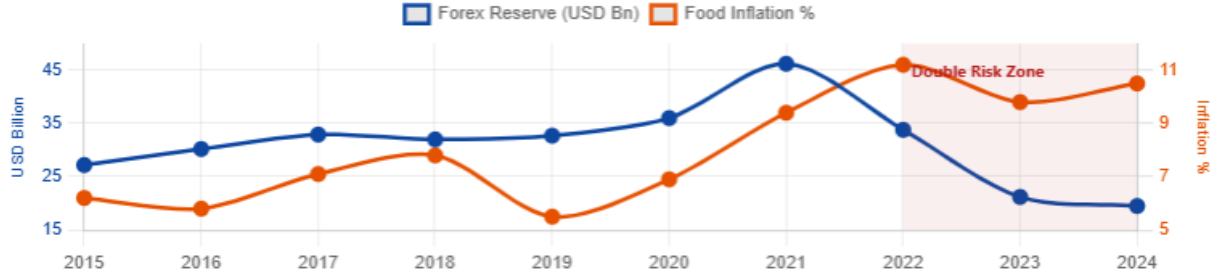
Global food prices are shock-driven, not stable. FAO vegetable oil index almost doubled to 171 in 2022. Every spike directly hits Bangladesh's import bill.

Export Ban Price Impact on Bangladesh (%)



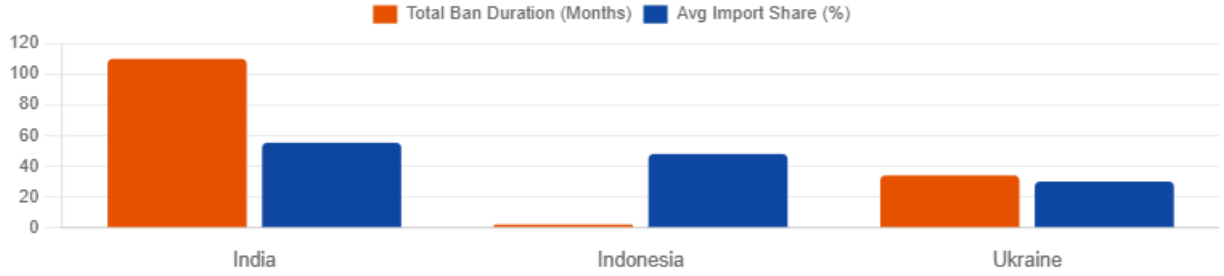
Export bans cause immediate domestic price surges. Indonesia's 2-month palm oil ban caused a 42% price rise. Bangladesh has no control over these external decisions.

Bangladesh Forex Reserve vs Food Inflation (%)



As forex reserves declined, food inflation rose sharply. Bangladesh's double vulnerability — high dependency and low reserves — makes every future shock more painful.

Export Ban Duration and Import Share by Country



India and Indonesia together supply the majority of Bangladesh's most critical food imports and have both imposed export bans affecting Bangladesh severely.

KrishiGrid Bangladesh

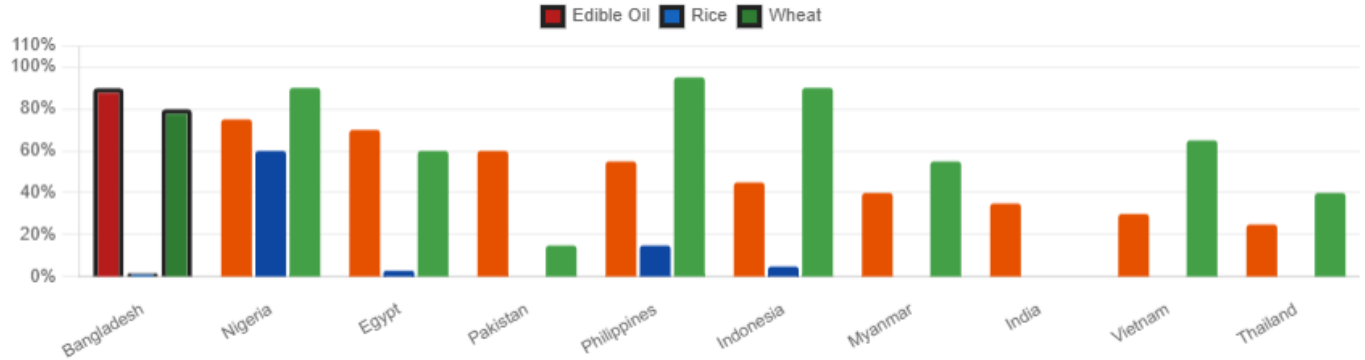
Bangladesh Oil Dependency
90%
% (Highest among peers)

Bangladesh Food Security Rank
84
out of 113 countries

Bangladesh Forex Cover
2.2
Months (vs India 11.5)

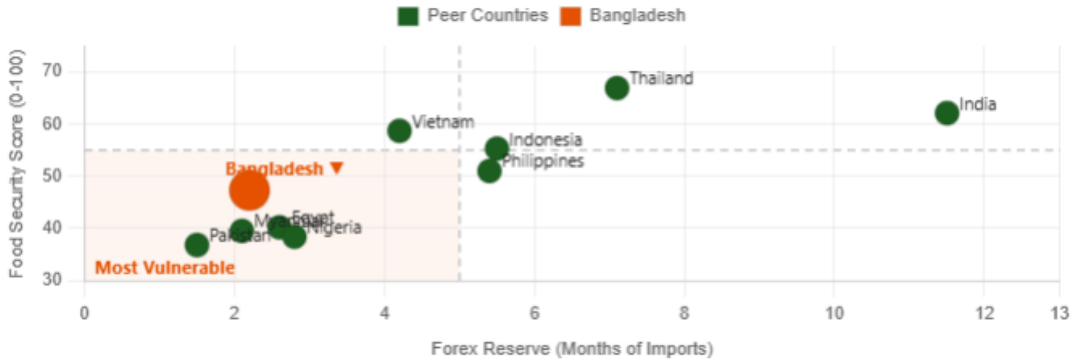
Best Peer Country Rank
#39
Thailand (Best in peer group)

Food Import Dependency by Country (%)



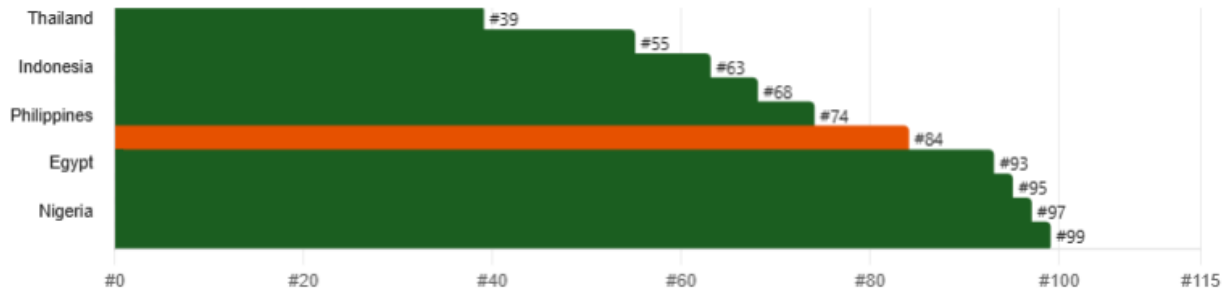
Bangladesh has the highest edible oil dependency (90%) among 10 comparable nations. Thailand and Vietnam reduced dependency through domestic programs — Bangladesh has not.

Food Security Score vs Forex Reserve Cover



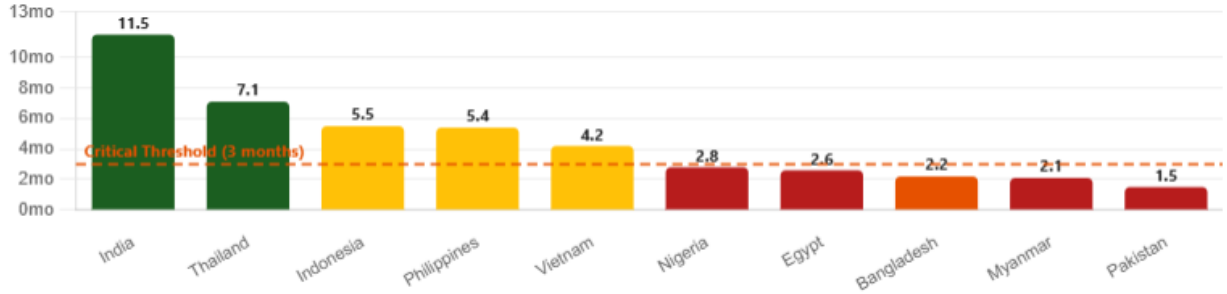
Bangladesh sits in the most vulnerable quadrant — low food security score and low forex buffer. Any future global shock hits Bangladesh harder than comparable countries.

Global Food Security Index Rank (Lower = Better)



Bangladesh ranks 84th in food security, far below Thailand (39) and Vietnam (55). The gap is driven by structural import dependency, not agricultural capability.

Forex Reserve Coverage (Months of Food Imports)



Bangladesh's 2.2 months of forex coverage is critically low. India maintains 11.5 months, Thailand 7 months. Bangladesh needs structural food production reform urgently.

KEY INSIGHTS

1. Global food markets are no longer stable systems but shock-driven environments, where prices spike sharply during crises rather than changing gradually

2. FAO indices and commodity prices show extreme volatility, with key items like vegetable oil and wheat doubling during global disruptions

3. These price shocks are directly linked to major global events such as COVID-19, wars, and climate impacts, making volatility a recurring pattern

4. Export bans by major countries like India and Indonesia prove that global supply cannot be relied upon during crises

5. Bangladesh's high import dependency (up to 90% in edible oil) makes it highly exposed to these external supply shocks

6. Price impacts from export bans are severe, with increases reaching up to 180% in essential commodities

7. Supply disruptions are often prolonged, lasting months to years, turning short-term policies into long-term constraints

8. Bangladesh has limited control over these risks, as they are driven by domestic priorities of exporting countries

9. Compared to peer countries, Bangladesh ranks lower in food security while maintaining higher dependency levels

10. Low forex reserves (≈2 months of import cover) further reduce the country's ability to absorb global shocks

11. Countries with stronger domestic production and reserves show significantly higher resilience during crises

12. Therefore, reducing import dependency is not optional it is a national security necessity to protect against future global food shocks